CS 335 - Fall 2026: Assignment 3

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Due: Monday November 16, 2026, 5pm

Submit all components of your solutions (written/analytical work, code/scripts/notebooks, figures, plots, etc.) to Crowdmark in PDF form in the section for each question.
You must also separately submit a single zip file containing any and all code/notebooks/scripts you write to the DropBox on LEARN in runnable format (that is, the files in the zip should be of type .ipynb).
For full marks, you must show your work and adequately explain your logic!

1. **(6 marks)** Floating Point Basics

   Consider a floating-point number system F specified by the parameters $\{\beta, t, L, U\}$, with base $\beta = 10$, t mantissa digits, and the exponent range given by $L = -8$ and $U = 8$. Under this system, a non-zero floating point number has the normalized form

   $$x = \pm 0.d_1 d_2 \cdots d_t \times \beta^p,$$

   with $d_1 \neq 0$.

   (a) (2 marks) The distance between 0.04 and the smallest representable floating point number larger than 0.04 is 10^{-7}. What is the smallest floating point number in F larger than 6000? Express your final answer in the normalized form specified above.

   (b) (2 marks) What are the smallest and largest *positive* numbers representable in F?

   (c) (2 marks) What is the value of machine epsilon, assuming rounding to the nearest value is used?

2. **(8 marks)** Effects of Cancellation

   The roots of a quadratic equation, $ax^2 + bx + c = 0$, are given by

   $$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \quad , \qquad x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

   when $a \neq 0$.

   (a) (3 marks) Consider performing arithmetic in a floating point number system with $\beta = 10$, $t = 6$, $L = -10$, $U = 10$ **with truncation**. Using the above formulas, you are to simulate by hand (with the aid of a calculator) the computation of the roots of the following quadratic equation

   $$0.3x^2 + 10.6x + 1 = 0, \tag{1}$$

   under this floating point system. (Obey the usual order of operations.) The true roots of the given quadratic, correct to six significant digits, are -0.0945929 and -35.2387. What is the *relative* error of each of your computed roots, x_1 and x_2?

(b) (2 marks) A *cancellation problem* arises when applying the above formulae for any quadratic equation having the property that

$$|b| \approx \sqrt{b^2 - 4ac} \; .$$

When this property holds, the quadratic formula for x_1 will exhibit cancellation if $b > 0$, while x_2 will exhibit cancellation if $b < 0$. We will try to circumvent this problem by reformulating our expression. Show that the following formula for x_1,

$$x_1 = \frac{2c}{-b - \sqrt{b^2 - 4ac}} \; ,$$

is mathematically equivalent to the usual one given at the start of the question (i.e., assuming you are working with real numbers rather than floating point numbers.) Hint: Rationalize the denominator of this expression.

(c) (1 mark) The formula for x_2 can be manipulated similarly. Deduce a better algorithm for calculating the roots of a quadratic equation (which minimizes cancellation errors), and present it in the form below (i.e., filling in the blanks for the x_1 and x_2 expressions).

Algorithm R.

if $b > 0$ then
    $x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$
    $x_1 = \frac{c}{a x_2}$
else
    $x_1 =$ ____
    $x_2 =$ ____
end

(d) (2 marks) Redo the calculation of the roots of equation (1) by applying Algorithm R using the same floating point number system. Compute the relative error of the computed roots and compare against your results from part (a). What do you observe?

3. **(8 marks)** Floating Point Error Analysis

Show that if x is a representable number in some floating point system F, the relative error of the floating point calculation $(x \otimes x) \ominus 1$ compared to the true value of $x^2 - 1$ is less than or equal to

$$\left(\frac{|x^2|}{|x^2 - 1|} + 1 \right) E \tag{2}$$

assuming we can drop/ignore terms proportional to E^2 and higher powers of E, and no overflow/underflow occurs. (Recall that the circle notation indicates a floating point counterpart of a corresponding real arithmetic operator.)

4. **(6 marks)** Convergence of Fixed Point Iteration

Consider a fixed point iteration:

$$x_{k+1} = g(x_k) \tag{3}$$

$$g(x) = \frac{\sin(x + \pi/2)}{2} \tag{4}$$

Will this iteration converge for any $x \in [0, \pi/2]$? Justify your answer by considering all the conditions of a convergence theorem.

5. **(16 marks)** Root finding methods

   In this question you will implement each of the four root finding algorithms discussed in class in a Jupyter/Python notebook. The function prototypes are as follows:

   - Bisection method.

     ```
     def bisection(f, a, b, tol, maxiter):
     % This function computes a root of f in the interval [a, b]
     % using the bisection method.
     ```

   - Secant method.

     ```
     def secant(f, a, b, tol, maxiter):
     % This function computes a root of f using the secant method
     % with initial guesses x_{-1} = a, x0 = b.
     ```

   - Newton's method.

     ```
     def newton(f, fprime, x0, tol, maxiter):
     % This function computes a root of f using the Newton's method
     % with initial guess x0. fprime should be the derivative of f.
     ```

   - Fixed point iteration.

     ```
     def fixedpoint(g, x0, tol, maxiter):
     % This function computes a fixed point of g using fixed point
     % iteration with initial guess x0.
     ```

   For each of these functions, the output should be a vector:

   $$x = [x_1, x_2, \ldots, x_n],$$

   where x_k is the approximate solution found after k iterations. The stopping criterion is when either $|x_n - x_{n-1}| < tol$ or the number of iteration exceeds the maximum, *maxiter*. You will also need to define the functions `f, fprime, g` according to the problem to be solved, below. (Note that Python allows you to pass functions as inputs to other functions, as implied by the function definitions above.)

   Find the root ($x^* = \sqrt{3}$) of the following equation:

   $$x^2 - 3 = 0, \qquad x \in [0.5, 3.5]$$

   using your four root finding functions with `tol=1e-10` and `maxiter=100`. For the bisection and secant methods, use `[a,b]=[0.5,3.5]`. For Newton and fixed point iteration, use `x0=0.5`. You will need to determine a $g(x)$ function such that the fixed point iteration converges. There are multiple valid choices here. (Do *not* use the $g(x)$ that corresponds to Newton's method.)

Next, using the root-finding methods above, write code that, for each of the methods, computes the errors of approximations, $e_k = |x_k - x^*|$, and generates a simple text-format table like the following:

Iteration	x_k	e_k	e_k/e_{k-1}	$e_k/e_{k-1}^{(1+\sqrt{5})/2}$	e_k/e_{k-1}^2
1	x_1	e_1	-	-	-
2	x_2	e_2	e_2/e_1	$e_2/e_1^{(1+\sqrt{5})/2}$	e_2/e_1^2
3	x_3	e_3	e_3/e_2	$e_3/e_2^{(1+\sqrt{5})/2}$	e_3/e_2^2
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Print the x, e, and various ratio values in scientific notation. A straightforward way to perform this type of formatting is using the `print` command, such as in the following (generic) example:

```
print("Row %d First Number:  %0.7f Second Number:  %0.7e" % (3, 1.73648392e-8, 1.73648392e-8))
```

which prints

```
Row 3 First Number:  0.0000000 Second Number:  1.7364839e-08
```

The %d is used to indicate an integer. %0.7f and %0.7e are used to display floating point numbers in standard and scientific notation, respectively, with the integer 7 here indicating the number of digits desired after the decimal. You can also use '\t' to insert tabs rather than spaces, such as `print("Word1\tWord2")`, which may help in aligning text columns. For the last 3 column headers, you can use the simpler titles 'ratio1', 'ratio2', and 'ratio3' in your code, rather than the mathematical expressions shown in the table above.

Using these tables, determine and state the order of convergence for each of the methods. Provide a brief text explanation of how you arrived at your answers.